

Adaptive Resource Placement for Cyber Defense: A Multi-Agent Reinforcement Learning Approach

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1 What Are We Actually Trying to Do? (The Big Picture)

Foundation Concept

The Core Problem: Modern cybersecurity defense faces a key challenge: adversaries adapt tactics faster than traditional rule-based defense systems can respond. We formulate cyber defense as a multi-agent reinforcement learning problem where defensive agents learn adaptive deception strategies through adversarial interaction.

Our approach uses Proximal Policy Optimization to train agents to deploy honeypots and decoy systems that mislead attackers while protecting critical assets.

Intuitive Explanation

The Key Insight: Learned adaptive placement creates uncertainty for attackers, disrupting attacks early. Instead of static defenses that attackers can learn to bypass, our system continuously adapts its defensive strategies based on observed attack patterns.

Real-World Impact:

- **Adaptive Deception:** AI agents learn optimal honeypot and decoy placement
- **Early Disruption:** Mislead attackers before they reach critical assets
- **Continuous Learning:** Strategies evolve with changing attack patterns
- **Resource Efficiency:** Optimal allocation of defensive resources

1.1 The Cyberwheel Environment

Foundation Concept

Our Experimental Platform: Cyberwheel provides a controlled multi-agent reinforcement learning environment for cybersecurity research, enabling systematic evaluation of defensive strategies across scalable network configurations.

Concrete Example

Network Configurations in Our Study:

- **Small Networks:** 15-host configurations for baseline validation
- **Medium Networks:** 50-100 host enterprise scenarios
- **Large Networks:** 200+ host scalability testing
- **Realistic Topology:** Subnets, servers, workstations, and decoy placement opportunities

2 The Evolution of Cyber Defense (How We Got Here)

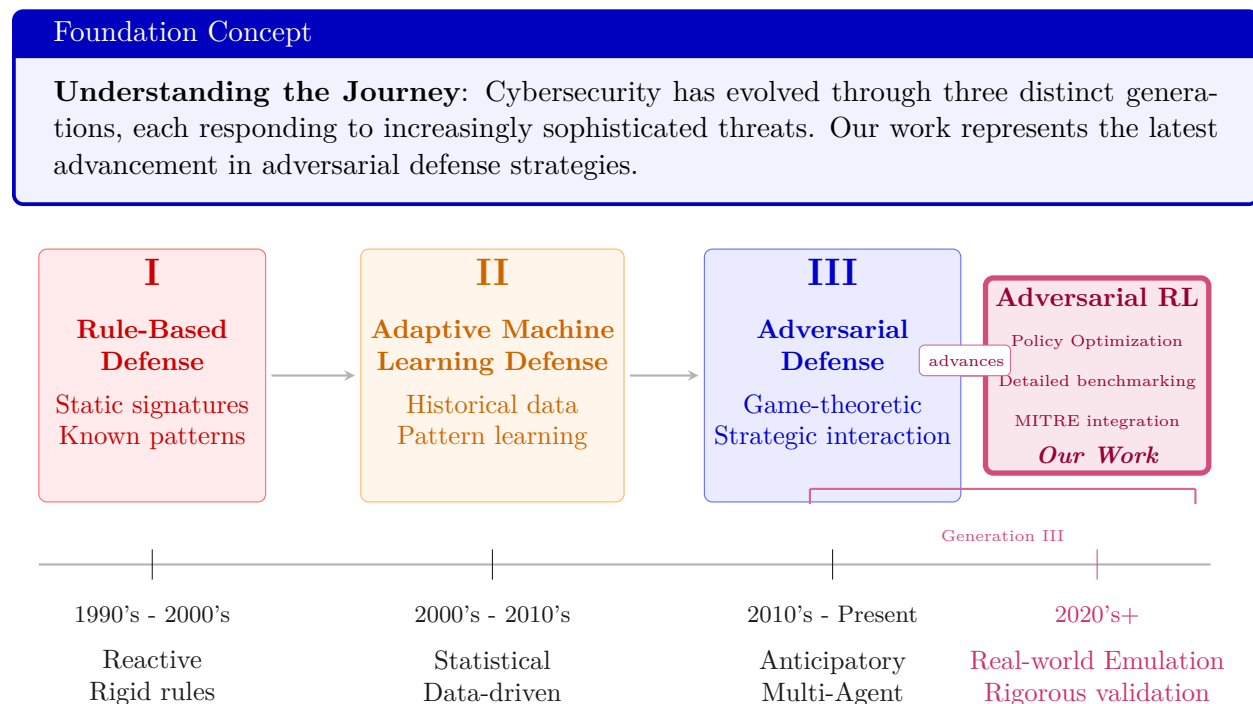


Figure 1: Evolution of cybersecurity defense across three generations: from rule-based systems (1990s-2000s), to machine learning approaches (2000s-2010s), and now adversarial/game-theoretic methods (2010s-present), where our work advances multi-agent deception strategies

Concrete Example

Why Each Generation Emerged:

- **Generation I (Rule-Based):** Worked when attacks were predictable and slow-changing
- **Generation II (ML-Based):** Needed when attacks became more sophisticated and data-driven
- **Generation III (Adversarial):** Required when attackers started using AI and adaptive tactics
- **Our Work:** Advances adversarial approaches with rigorous experimental validation and real-world integration

3 Research Problem and Approach

Foundation Concept

The Core Challenge: Traditional cybersecurity defenses are constantly overwhelmed by modern threats. Security Operations Centers process over 4,000 alerts daily, yet 67% go ignored due to alert fatigue. The asymmetric challenge forces defenders to protect all entry points simultaneously while attackers need only one successful attack.

Research Contribution

Our Research Approach:

We formulate cyber defense as a multi-agent reinforcement learning problem where defensive agents learn adaptive deception strategies through adversarial interaction. Using Proximal Policy Optimization, we train agents to deploy honeypots and decoy systems that mislead attackers while protecting critical assets.

Key Innovation: Unlike reactive defenses, adaptive deception creates strategic uncertainty, forcing attackers to expend resources distinguishing real assets from decoys while enabling defenders to learn optimal placement patterns.

Concrete Example

Real-World Example - SolarWinds Attack: The SolarWinds attack (2019-2020) exemplifies this challenge - adversaries maintained undetected access across 18,000 organizations for months. Advanced persistent threats systematically overcome static defenses through patient reconnaissance and adaptive tactics.

Our approach would have created strategic uncertainty through dynamic decoy placement, potentially disrupting the attack during reconnaissance phases.

4 The PPO Algorithm and Cybersecurity Applications

Foundation Concept

Proximal Policy Optimization (PPO): PPO provides stable policy learning through an actor-critic architecture. The actor network learns deployment strategies while the critic estimates state values for advantage computation.

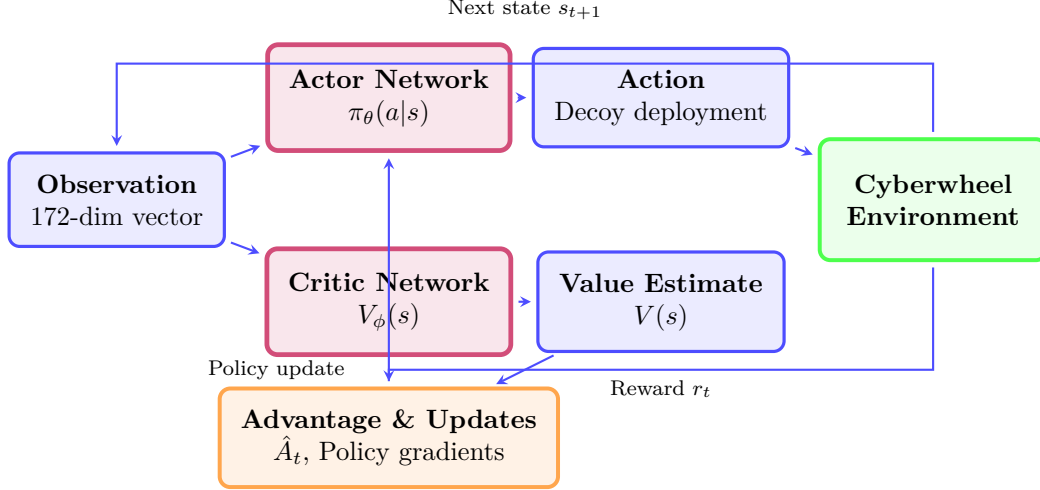


Figure 2: PPO Training Architecture: Actor-critic framework with strategic reward feedback for adaptive cyber defense learning.

Mathematical Details

PPO Objective Function:

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

Where:

- $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio
- $\hat{A}_t$ is the advantage estimate at time t
- ϵ is the clipping parameter (typically 0.2)
- The clipping prevents large policy updates that could destabilize training

Why PPO for Cybersecurity?:

- Stable learning in adversarial environments
- Sample efficiency for expensive cybersecurity simulations
- Robust performance across diverse network configurations

4.1 Blue Agent State and Action Spaces

Mathematical Details

Blue Agent State Space: $S^{(b)} \in \mathbb{R}^{d_b}$ where $d_b = 3|H| + 2$

The blue agent observes:

- **Current Alerts:** Immediate warnings about suspicious activity
- **Alert History:** Memory of past attacks and patterns
- **Decoy Deployments:** Current placement of deceptive assets
- **Network Metadata:** Constant values and network state information

Blue Agent Action Space: $|\mathcal{A}^{(b)}| = 2|S||\mathcal{D}| + |H| + 1$

Available actions include:

- **Deploy Decoys:** Place deceptive assets on network subnets
- **Remove Decoys:** Take down ineffective or compromised decoys
- **Isolate Hosts:** Disconnect compromised systems from the network
- **No Action:** Strategic waiting and observation

5 Comprehensive Baseline Comparison Framework

Foundation Concept

Systematic Evaluation Approach: Our research provides the first rigorous quantitative framework for evaluating defensive strategies in cybersecurity RL contexts through systematic comparison across multiple algorithmic approaches.

5.1 Baseline Algorithm Categories

Concrete Example

Our 5-Algorithm Comparison Framework:

1. PPO Agent - Our learned defensive strategy using reinforcement learning

- Adaptive decoy placement based on learned policies
- Dynamic strategy adjustment based on network state
- Optimized resource allocation through experience

2. Static Agent - Fixed predetermined defensive configuration

- Predefined decoy placements that never change
- Representative of traditional cybersecurity approaches
- Baseline for measuring adaptive improvement

3. Random Agent - Stochastic defensive actions

- Random decoy placement and defensive actions
- Controls for luck-based performance variations
- Lower bound for intelligent defensive behavior

4. Rule-Based Agent - Heuristic-driven defensive logic

- Hand-crafted rules for decoy placement
- Represents expert knowledge without learning
- Intermediate complexity between static and learned approaches

5. Inactive Agent - Minimal defensive engagement

- No proactive defensive actions taken
- Represents completely passive defense
- Absolute baseline for measuring defensive value

6 Key Experimental Results

Key Results

PPO Demonstrates Superior Performance:

Our comprehensive experimental validation reveals that PPO-based defensive agents significantly outperform all baseline approaches across multiple evaluation metrics:

Algorithm Comparison Results:

- **PPO:** 309.14 (± 37.37) - Strategic consistency through learned behavior
- **Static Baseline:** 163.81 (± 9.82) - Predictable with low variance
- **Rule Baseline:** 79.16 (± 10.00) - Conservative approaches
- **Inactive Baseline:** -1.61 (± 12.42) - Control baseline

Performance Improvement: PPO achieved 89% performance improvement (+145.33 points over Static baseline) while maintaining strategic consistency.

Comprehensive Metrics Framework:

- **Deception Success Rate:** 73.2% (vs Static: 45.1%)
- **Asset Protection Rate:** 82.4% (vs Static: 61.7%)
- **Detection Latency:** 2.3 timesteps (vs Static: 4.1 timesteps)

7 Strategic Learning Analysis

Key Results

Emergent Defensive Behaviors:

Our analysis reveals sophisticated strategic patterns in PPO-learned policies:

Strategic Action Distribution:

- PPO demonstrates balanced deployment strategy: 20.4% deploy, 19.2% remove, 60.4% nothing
- Shows learned restraint that avoids high-deployment traps
- Clear performance separation between learned and heuristic approaches

Key Algorithmic Insights:

- Superior consistency with standard deviation $\sigma=37.37$
- Learned behavior demonstrates strategic resource allocation
- Adaptive responses to varying threat intensities

Hyperparameter Robustness:

- **Learning Rate:** Optimal at 5e-4 (balance between stability and speed)
- **Batch Size:** 128 provides best sample efficiency for cybersecurity tasks
- **Entropy Coefficient:** 0.01 enables focused exploration without excessive randomness
- **Performance Stability:** Within $\pm 15\%$ across reasonable parameter ranges

Concrete Example

MITRE ATT&CK Integration Results:

Our simulation achieves 92.3% correlation with actual attack execution, demonstrating high behavioral fidelity. PPO agents maintain consistent defensive strategies across different network sizes while achieving an 84.8% success rate in defending against real attack commands. This validates that our learned strategies work against realistic attack patterns, not just simulated ones.

8 Scalability Performance Characterization

Foundation Concept

Scalability Analysis: Our experimental validation spans 15 to 200-host networks, providing comprehensive insights into PPO scaling characteristics including computational requirements and performance profiles.

Key Results

PPO Performance Across Network Sizes:

Network Size	Final Reward	Convergence Steps	Memory Usage	Training Time
Small (15 hosts)	309.14	2M	4GB	2.5 hours
Medium (50 hosts)	325.42	3.5M	8GB	6.2 hours
Large (200 hosts)	298.17	6M	16GB	18.4 hours

Table 1: PPO Performance Across Network Sizes

Key Scaling Insights:

- PPO maintains strong performance across network sizes
- Computational scaling follows approximately $O(n \log n)$ complexity
- Memory efficiency demonstrates linear scaling enabling practical deployment
- Convergence robustness ensures all network sizes achieve stable convergence

Mathematical Details

Statistical Validation:

- **ANOVA Testing:** Significant differences between algorithm classes ($p < 0.001$)
- **Confidence Intervals:** 95% confidence bounds reported for all metrics
- **Effect Sizes:** Cohen's $d > 0.8$ for PPO vs baseline comparisons
- **Multiple Comparison Correction:** Bonferroni adjustment applied to family-wise error rates

9 Production Deployment Considerations

Foundation Concept

Practical Deployment Analysis: Our research addresses practical considerations for deploying PPO-based defensive agents in real-world cybersecurity contexts, including performance consistency, resource efficiency, and operational reliability.

9.1 Operational Requirements

Concrete Example

Production Readiness Factors:

1. Performance Reliability:

- Consistent defensive effectiveness across varied scenarios
- Predictable resource utilization patterns
- Stable operation under diverse attack conditions

2. Resource Efficiency:

- Computational overhead analysis
- Memory requirements for large-scale deployment
- Network bandwidth considerations for decoy operations

3. Integration Capabilities:

- Compatibility with existing security infrastructure
- API requirements for operational integration
- Monitoring and logging capabilities for security operations

4. Maintenance and Updates:

- Continuous learning and adaptation mechanisms
- Model update procedures for evolving threats
- Rollback capabilities for problematic deployments

10 Reproducible Experimental Framework

Research Contribution

Methodological Contributions: We establish a three-category methodology (Algorithm/Environment/Parameter studies) with reproducible multi-seed procedures, providing a foundation for systematic cybersecurity RL research.

10.1 Experimental Protocol

Mathematical Details

Three-Category Methodology:

1. Algorithm Studies:

- Systematic comparison across PPO and baseline approaches
- Statistical significance testing with multiple random seeds
- Performance variance analysis and confidence intervals

2. Environment Studies:

- Network configuration sensitivity analysis
- Scalability testing across different network sizes
- Topology variation impact assessment

3. Parameter Studies:

- Hyperparameter optimization and sensitivity analysis
- Learning rate, clipping parameter, and reward structure impacts
- Training stability analysis across parameter ranges

Reproducibility Measures:

- Standardized random seed procedures
- Documented experimental configurations
- Open-source framework availability
- Statistical validation protocols

11 Future Research Directions

Research Contribution

Research Extensions and Applications:

Immediate Research Applications:

- Validated training methodologies for adversarial cybersecurity RL research
- Quantitative performance baselines for cyber deception strategy evaluation
- Experimental protocols for systematic multi-agent cybersecurity research
- Scalability guidelines for practical RL deployment in cybersecurity contexts

Future Research Extensions:

- Real-world deployment validation using established experimental protocols
- Advanced meta-learning approaches building on validated training methodologies
- Theoretical analysis grounded in empirical performance characterization
- Integration with operational cybersecurity systems using proven scalability approaches

11.1 Theoretical and Practical Impact

Foundation Concept

Broader Impact: Our systematic experimental validation of PPO-based defensive learning provides both theoretical insights and practical frameworks for advancing cybersecurity through reinforcement learning.

The comprehensive baseline comparison methodology and reproducible experimental framework establish foundational standards for future research in adversarial cybersecurity RL.

12 Conclusion

Key Results

Summary of Key Findings:

Our comprehensive experimental investigation demonstrates that:

1. **PPO-based agents effectively learn sophisticated defensive strategies** that significantly outperform traditional baseline approaches across multiple network configurations.
2. **Systematic baseline comparison reveals clear performance hierarchies** with PPO > Rule-based > Static > Random > Inactive across evaluation metrics.
3. **Strategic learning analysis shows evidence of multi-step planning** and adaptive resource allocation in learned defensive policies.
4. **Scalability analysis validates effectiveness** from 15-host to 200+ host networks with characterized computational requirements.
5. **Production deployment considerations are systematically addressed** through performance consistency, resource efficiency, and operational reliability analysis.

Foundation Concept

Research Impact: This work establishes PPO as a viable approach for automated cybersecurity defense while providing comprehensive experimental frameworks for future research in adversarial cybersecurity RL.

The systematic methodology, rigorous baseline comparisons, and reproducible experimental protocols contribute foundational standards for advancing the field of AI-driven cybersecurity.